QUALITY ENGINEERING · DATA INTEGRATION

Banking Data Warehouse ETL Test Strategy

End-to-end validation of customer, account, transaction and branch data loads

DRAFT — PENDING HUMAN REVIEW

Project	Banking Data Warehouse ETL (BRD project: Banking ETL Testing)
Target application	SQL Server Data Warehouse
Document version	1.0 — Initial draft
Prepared by	QA / Test Strategy — author to confirm
Prepared date	02-Nov-2026
Requirements source	Business_Requirements/BRD_Banking_ETL_v1.md, version 1.0

This is a review draft, not an approval or evidence that the ETL implementation, mappings, environments, or controls have been verified.

1. Document Control

Version	Date	Author	Description / status
1.0	02-Nov-2026	QA / Test Strategy (to confirm)	Initial project-specific draft; pending human review and approval.

Source reviewed: Banking Data Warehouse — Business Requirements Document, version 1.0. The supplied example was used for structure and presentation only; its sample systems, tools, roles, and values are not treated as project facts.

2. Introduction and Objectives

Purpose

Define a risk-based, requirements-led strategy to validate the end-to-end ETL flow that integrates customer, account, transaction, and branch information into the SQL Server Data Warehouse. This draft establishes coverage, evidence, entry/exit controls, and author questions; it does not define detailed test cases or unprovided business rules.

Objectives

- Verify that valid customer, account, transaction, and branch data is loaded to the intended warehouse structures.
- Validate the stated account-to-customer, transaction-to-account, and branch-master relationships.
- Check source-file structure, data quality, transformation accuracy, completeness, and source/target reconciliation.
- Verify the agreed treatment of duplicates and invalid records, including error/reject handling and execution logs.
- Exercise incremental-change processing, package failure recovery, and restartability against approved rules.
- Provide traceable execution evidence and surface unresolved requirement risks before approval.

3. Project Overview

The BRD describes a banking data warehouse intended to centralize customer, account, transaction, and branch reference information for business reporting, transaction analysis, and customer insights. The required target is SQL Server, with dimensions for customer, account, and branch and a transaction fact table.

> **Requirement maturity:** The BRD is high-level. It states that detailed transformation rules, rejection criteria, and expected results will be finalized in source-to-target mapping and business-rule documents. Those documents were not present in the supplied project files; detailed expected results therefore remain open.

4. ETL Architecture and Data Flow

The following logical flow is supported by the BRD. Physical architecture and implementation details must be confirmed before execution.

Stage / component	Confirmed information	Not confirmed
Input	customers.csv, accounts.csv, transactions.csv, and branch_master.xlsx.	File locations, delivery mechanism, encoding, delimiters, header conventions, file naming/versioning, and arrival schedule.
Ingestion / staging	ETL must read the source files.	ETL product, staging area/schema, landing/archive behavior, orchestration, and technical design.
Transform / validate	Validate required relationships, branch data, duplicates, invalid records, and transaction amount accuracy.	Mapping, transformations, keys, reject criteria, duplicate policy, and data-quality thresholds.
Warehouse target	SQL Server; Dim_Customer, Dim_Account, Dim_Branch, Fact_Transaction.	Database/schema names, table definitions, constraints, surrogate-key policy, and retention/history behavior.
Operations	Package failures and rejected records must be logged; incremental changes must be processed.	Scheduler, logging schema, alerting, retry/restart controls, watermark strategy, and SLAs.

5. Testing Scope

In scope

- CSV/Excel file presence, readability, structure, headers, column types/formats, and malformed-file handling.
- Source profiling and validation of required fields, nulls, duplicates, invalid values, and boundary values, once rules are approved.
- Source-to-target mapping, transformations, data types, precision/scale, keys, and relationship integrity.
- ETL package/pipeline execution, row routing, error logging, rejected-record handling, failure recovery, and restartability.
- Full and incremental loads, repeat-run/idempotency behavior, source-to-target reconciliation, regression, and negative testing.
- Business checks for transaction amount accuracy and the account/customer, transaction/account, and branch-master relationships.

Out of scope unless brought into the approved requirements

- Functional testing of upstream banking applications and downstream reporting/UI behavior.
- Infrastructure, disaster recovery, penetration/security, performance/load, availability, accessibility, and localization testing as formal test phases; related non-functional requirements are currently unspecified and must be assessed.
- Validation of business definitions or financial calculations not documented in approved mapping/business rules.

6. Source and Target Systems

Source	Format / content	Target	Relationship / rule from BRD
customers.csv	CSV — customer details	SQL Server / Dim_Customer	All valid customer records must be loaded.
accounts.csv	CSV — bank account details	SQL Server / Dim_Account	Each account must belong to a valid customer.

Source	Format / content	Target	Relationship / rule from BRD
transactions.csv	CSV — financial transactions	SQL Server / Fact_Transaction	Each transaction must reference a valid account; amount must load accurately.
branch_master.xlsx	Excel — branch reference information	SQL Server / Dim_Branch	Branch information must be validated against the branch master.

The BRD names these source files and target tables but does not supply their schemas, key columns, complete mapping, or exact branch-reference matching rules.

7. Testing Methodology

1. Baseline the approved BRD, source-to-target mapping (STTM), business rules, target schema, and operational design; record unresolved requirements.
2. Review source layouts and profile representative data; agree test datasets, privacy controls, and expected outcomes with data owners.
3. Derive traceable scenarios from approved requirements and rules. Classify each as confirmed coverage or blocked pending clarification.
4. Validate environment access, file delivery, database connectivity, logging, and baseline target state.
5. Execute source/file checks, package/pipeline checks, transformation and relationship checks, full-load reconciliation, then incremental, recovery, negative, and regression tests.
6. Compare outcomes with approved expected results, preserve SQL/query outputs and run logs, and log defects with reproducible evidence.
7. Retest fixes, assess regression impact, report residual risks and exceptions, and obtain required business/technical sign-off.

> **Traceability rule:** Every pass/fail criterion must come from an approved requirement, mapping, or business rule. Do not invent tolerances, rejection thresholds, duplicate winners, or incremental semantics where the source documents are silent.

8. ETL Testing Types and Coverage

Test type	Coverage focus	Trace
File / schema validation	Expected file set, readable files, worksheet/header/column layout, required columns, format/type/length, malformed or missing input behavior.	BRD §2; clarification needed for file contract.
Data quality / negative	Null, malformed, out-of-domain, duplicate, orphan, empty-file, and boundary records; verify reject/quarantine and logs per approved policy.	BR-05, BR-06; criteria TBD.
Mapping / transformation	Column mapping, derivations, conversions, precision/scale, defaults, and transaction amount preservation/rounding.	BR-07; STTM and rules missing.
Referential integrity	Account has valid customer; transaction references valid account; branch values are validated against branch master.	BR-02, BR-03, BR-04.
Completeness / reconciliation	Source-to-target counts, accepted/rejected totals, and agreed field-level/aggregate comparisons by entity and load.	BR-08; reconciliation grain/tolerance TBD.

Test type	Coverage focus	Trace
Incremental load	New/changed records, unchanged reruns, late arrivals, repeat processing, and duplicate prevention according to the agreed change-detection rules.	BR-09; watermark/change rules TBD.
Package / operational	Successful run, dependency/connectivity failure, reject logging, failure recovery, restartability, and execution evidence.	BR-10; operational design TBD.
Regression	Re-run impacted schema, transformation, relationship, reconciliation, and incremental coverage after a change/fix.	Approved baseline and change impact.

9. ETL Package and Pipeline Testing

- Verify configured source and target connectivity, package parameters, file selection, and execution outcome against the approved design.
- Check component-level row counts or equivalent available run metrics at ingestion, transform, load, and reject boundaries.
- Verify failures are surfaced and logged with enough run, file, step, and error context to diagnose the issue; confirm rejected rows are traceable.
- Test recovery and restart from agreed failure points, including repeat execution, without unintended duplicate or partial target data.
- Verify run logs and final status agree with observed target and reject outcomes. Exact log fields, retry behavior, and alerting remain to be agreed.

10. Source-to-Target Validation and Data Reconciliation

Use approved mappings and keys to compare each source entity with its target. Capture the input population, accepted population, rejects, and target population for the same execution/batch.

- Compare source, accepted, rejected, and target record counts; reconcile counts using an agreed formula and treatment of duplicates/rejections.
- Compare mapped columns and approved transformations for representative records and boundary values; verify data types, precision, and truncation/rounding rules.
- Validate customer/account and account/transaction relationships and branch-master matching against confirmed business keys.
- Compare transaction amount totals and any other agreed aggregates by an approved grouping and currency/rounding rule; do not assume amount sign or currency semantics.
- Check duplicate, null, and invalid-record outcomes against the agreed disposition; reconcile error-table/reject counts to logged counts.
- For incremental runs, compare expected inserts/updates/no-change rows to the agreed change set and verify rerun behavior.

> **Acceptance thresholds:** No numeric reconciliation tolerance, allowable reject rate, or exact comparison key is specified in the BRD. These must be agreed before pass/fail reconciliation can be approved.

11. Test Environment, Tools and Test Data

Need	Current status	Strategy / decision required
ETL runtime and scheduler	Not identified.	Confirm tool, version, package/pipeline inventory, parameters, and how runs are triggered.
SQL Server test target	SQL Server is the target technology; endpoint/version/schema not supplied.	Provide isolated test database, schema, connectivity, access roles, and refresh/reset procedure.
Source/test files	Four filenames/formats are named; actual sample files and contracts were not supplied.	Provide representative and controlled negative/boundary fixtures; protect sensitive banking/customer data.
Validation and evidence	Specific tools are not prescribed.	Use approved SQL client/query method, ETL run logs, controlled evidence storage, and agreed defect tracker.

Test data should include valid linked entities, missing/invalid references, nulls, duplicates, empty and malformed files, numeric/date boundaries, transaction amount precision cases, and incremental insert/change/replay cases. Values and expected outcomes must be authorized by the data owner and aligned to approved rules. Use masked or synthetic data unless approved access to production-derived data is explicitly granted.

12. Defect Management

Record defects in the project-approved issue/defect tracker (tool TBD). Each record should include requirement or gap reference, environment/build, source file and batch/run identifier, reproducible steps, test data reference, expected and actual results, severity/priority, relevant query/log evidence, owner, status, and retest/regression result. Protect sensitive data in screenshots, logs, attachments, and SQL output. Severity definitions, triage cadence, and escalation contacts require project agreement.

13. Roles and Responsibilities

Role	Proposed responsibility	Assignment
QA / Test Lead	Own strategy, coverage, execution coordination, risk/defect reporting, and test summary.	To be assigned.
ETL Developer / Engineer	Explain implementation, support package execution, diagnose/fix defects, and provide technical logs.	To be assigned.
Data / Database Administrator	Provide target schema/access, environment support, refresh, and approved data extracts.	To be assigned.
Business Analyst / Data Owner	Confirm mappings, business rules, rejection/duplicate policy, and expected results.	To be assigned.
Business Owner / Approver	Review residual risks, accept business exceptions, and approve strategy/exit where authorized.	To be assigned.

Roles are proposed from the example strategy's generic role model, not confirmed project staffing.

14. Entry and Exit Criteria

Entry criteria

- BRD reviewed and open questions triaged; approved STTM, business rules, target schema, reject/duplicate policy, and incremental design available for the test scope.
- ETL build and deployable test version identified; environment, connectivity, access, logging, and controlled source/test data are ready.
- Traceable scenarios and expected outcomes reviewed; test reset and evidence handling agreed.

Exit criteria

- Approved planned coverage has been executed and results/evidence recorded; blocked, failed, and not-run items are explicitly reported.
- Count, data, relationship, reject, and incremental reconciliation is completed against approved thresholds, or documented exceptions are accepted by the authorized owner.
- No unresolved critical/high defects remain unless formally risk-accepted; fixes have been retested and impacted regression has passed.
- Execution summary, residual risks, known limitations, and required business/technical approvals are recorded.

Final numeric completion thresholds, severity gates, and approvers are TBD; this draft does not claim that any exit criteria have been met.

15. Risks and Dependencies

Risk / dependency	Impact	Mitigation / action
STTM, detailed business rules, rejection criteria, and expected results are unavailable.	Transformation and reject outcomes cannot be judged reliably; strategy approval and execution are blocked for affected coverage.	Obtain owner-approved documents and resolve Section 16 questions before baselining expected results.
Duplicate and invalid-record disposition is undefined.	Count reconciliation and correctness may differ between valid interpretations.	Approve deterministic duplicate selection and reject/quarantine rules, including auditability.
Incremental change/watermark behavior is undefined.	Missed, repeated, or late changes may be undetected.	Document change keys, update rules, cutoff/watermark, replay, and late-arrival handling; test agreed cases.
File schemas, target schema, and test data/environment are not supplied.	File and database validation cannot be prepared or executed.	Publish versioned contracts, provision isolated environment/access, and provide representative controlled data.
Sensitive banking/customer data may be used.	Privacy or access-control exposure in test data/evidence.	Prefer masked/synthetic data; obtain authorization and apply least privilege and evidence redaction/retention rules.
No non-functional targets or operational SLAs are stated.	Performance, availability, security, and recovery adequacy cannot be assessed.	Business/operations owners to define applicable targets and whether additional test phases are required.

16. Deliverables and Approval

Deliverables

- This ETL Test Strategy (draft pending human review).
- After review/approval: traceable scenarios and test cases, controlled test-data inventory, execution evidence, defect records, reconciliation results, and test execution summary.
- Approved mapping, business-rule, test-environment, and run/logging references needed to execute and assess results.

Approval record

Approver role	Name	Decision / comments	Date
QA / Test Lead	TBD	Pending review	TBD
Business Owner / Data Owner	TBD	Pending review	TBD
ETL / Technical Owner	TBD	Pending review	TBD

Gaps and Questions for the Author

The following items were absent from the supplied BRD/project files. “Approval blocker” means the strategy cannot be baselined for the affected coverage until the decision or document is available.

#	Gap / impact	Question for author / owner	Approval impact
1	Source-to-target columns, keys, conversions, defaults, and derived fields are not defined; expected transformation results cannot be established.	Can you provide the approved STTM and target schemas for all four entities?	Blocker for transformation and reconciliation approval.
2	“Valid” customer/account/transaction and branch match criteria are unspecified.	What field-level validations, required fields, valid domains, key rules, and branch matching rules define a valid record?	Blocker for data-quality and relationship pass/fail.
3	Duplicate policy is not defined.	Should duplicates be rejected, deduplicated, or retained? What business key, precedence, and audit behavior apply?	Blocker for duplicate outcomes and count reconciliation.
4	Invalid-row disposition is stated as reject or error table but not selected or specified.	Which invalid records are rejected, where are they stored, what error details are required, and can a batch partially succeed?	Blocker for reject/error handling approval.
5	Incremental processing semantics are not described.	What identifies inserts/updates/deletes, how are watermarks advanced, and how are reruns, late-arriving rows, and corrections handled?	Blocker for incremental and restartability approval.
6	File contracts and delivery expectations are absent.	What are required columns, encoding/delimiter, worksheet/sheet name, naming/location, optionality, empty-file policy, and arrival schedule?	Blocker for file validation readiness; some cases can be planned later.

#	Gap / impact	Question for author / owner	Approval impact
7	Reconciliation grain and acceptable differences are unspecified.	Which keys/aggregates and batch boundaries must reconcile, how are rejects/duplicates counted, and what tolerance is allowed (especially for amounts)?	Blocker for reconciliation sign-off.
8	ETL tooling, architecture, environment, test database, and run access are unknown.	Which ETL/orchestration versions and environments are in scope; who provisions connectivity, access, reset, and logs?	Blocker to execution; strategy can be reviewed provisionally.
9	Privacy and security requirements for data and execution access are not stated.	What classification, masking, retention, access, and evidence-handling controls apply? Are security-specific tests required?	Approval needed before test data/access; security scope remains undecided.
10	Performance, availability, recovery objectives, and operational SLAs are absent.	Are volume, run-window, throughput, availability, retry, and recovery targets required, and who approves them?	Non-functional scope cannot be accepted until decided.
11	Defect tracker, severity/priority rules, triage, and named approvers are not identified.	Which tracker, triage cadence, exit thresholds, and authorized business/technical approvers should be used?	Needed before execution governance and final sign-off.
12	No detailed acceptance criteria or Jira reference was provided.	Are there approved acceptance criteria, release scope, or linked change tickets to include in traceability?	Approval blocker if additional scope/criteria exist.

Human Review Gate

Document status: DRAFT — PENDING HUMAN REVIEW. Not approved.

Information used: The supplied Banking Data Warehouse BRD v1.0, the supplied example test-strategy document for structure, and the Test Strategy Creator guidance. The BRD identifies the four input files, SQL Server target tables, high-level relationship/data-quality/incremental/logging requirements, and intended business outcome.

Assumptions made: No project behavior was added as a confirmed fact. Proposed test techniques, generic role responsibilities, and entry/exit controls are recommendations pending owner agreement. The cover's project label summarizes the BRD and is not a separately confirmed application name.

Could not be confirmed: Detailed mapping/business rules, schemas/keys, duplicate and rejection rules, incremental semantics, ETL tooling/architecture, environments, actual test data, reconciliation thresholds, non-functional targets, defect process, staffing, and approvers.

Approval blockers: At minimum, resolve Questions 1–8 sufficiently to baseline expected outcomes, reconciliation, and execution readiness; agree applicable data-protection and non-functional scope; and identify authorized approvers. See the gaps table for impacts and owners to consult.

Review request: QA, ETL, data, and business owners should correct project facts, answer the open questions, confirm scope and measurable exit thresholds, and explicitly approve or return this draft for revision. No detailed test cases, SQL validation scripts, or automation are included in this draft.

Human decision: ☐ Approve after updates ☐ Return with edits Reviewer: _____ Date: _____

End of draft — approval must be recorded by the authorized human reviewers before this strategy is baselined.